

Модуль 4. Лабораторная 2

1) Для заданной на схеме schema-lab2 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить протокол STP, назначив явно один из коммутаторов корневым настройкой приоритета.

```
enable
configure terminal

spanning-tree mode pvst
spanning-tree vlan 1 priority 4096

end
write memory
```

2) Проверить доступность каждого с каждым всех персональных компьютеров (VPCS), результаты запротоколировать.

```
Pc1: 192.168.1.1/24
Pc2: 192.168.1.2/24
Pc3: 192.168.1.3/24
Pc4: 192.168.1.4/24
Pc5: 192.168.1.5/24
Pc6: 192.168.1.6/24
```

PC1

```
PC1> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=9.760 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=10.927 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=6.717 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=8.502 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=11.410 ms
```

```
PC1> ping 192.168.1.3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=21.875 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=19.357 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=31.971 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=34.014 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=13.564 ms
```

```
PC1> ping 192.168.1.4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=27.122 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=20.346 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=32.051 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=13.722 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=45.993 ms
```

```
PC1> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=15.271 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=28.451 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=25.044 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=32.282 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=29.001 ms
```

```
PC1> ping 192.168.1.6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=26.351 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=31.836 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=22.867 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=32.638 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=14.773 ms
```

PC2

```
PC2> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=8.528 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=21.951 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=12.542 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=14.201 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=15.004 ms
```

```
PC2> ping 192.168.1.3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=13.182 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=36.509 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=14.507 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=16.247 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=43.867 ms
```

```
PC2> ping 192.168.1.4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=20.239 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=17.615 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=39.282 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=24.063 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=15.791 ms
```

```
PC2> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=18.394 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=11.356 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=26.689 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=39.720 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=24.802 ms
```

```
PC2> ping 192.168.1.6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=25.381 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=18.318 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=15.131 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=43.937 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=27.686 ms
```

PC3

```
PC3> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=34.622 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=22.836 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=27.244 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=34.114 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=14.061 ms
```

```
PC3> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=46.955 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=31.023 ms
```

```
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=22.501 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=20.599 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=22.436 ms
```

```
PC3> ping 192.168.1.4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=6.967 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=4.267 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=6.951 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=16.543 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=22.052 ms
```

```
PC3> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=23.584 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=13.928 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=26.011 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=11.169 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=17.261 ms
```

```
PC3> ping 192.168.1.6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=14.225 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=29.458 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=26.569 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=21.221 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=12.049 ms
```

PC4

```
PC4> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=30.891 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=37.190 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=20.463 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=20.981 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=26.292 ms
```

```
PC4> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=8.587 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=18.192 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=33.370 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=35.565 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=49.178 ms
```

```
PC4> ping 192.168.1.3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=6.080 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=16.165 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=5.306 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=10.158 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=7.624 ms
```

```
PC4> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=30.515 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=22.053 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=35.681 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=16.939 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=33.074 ms
```

```
PC4> ping 192.168.1.6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=19.125 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=39.212 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=13.260 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=16.006 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=41.488 ms
```

PC5

```
PC5> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=28.169 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=10.408 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=20.094 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=22.752 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=36.769 ms
```

```
PC5> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=16.013 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=40.667 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=32.685 ms
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=33.872 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=16.520 ms
```

```
PC5> ping 192.168.1.3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=13.409 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=18.605 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=16.409 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=28.769 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=31.353 ms
```

```
PC5> ping 192.168.1.4
```

```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=41.190 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=24.048 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=33.920 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=17.874 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=17.890 ms
```

```
PC5> ping 192.168.1.6
```

```
84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=8.092 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=4.190 ms
84 bytes from 192.168.1.6 icmp_seq=3 ttl=64 time=4.122 ms
84 bytes from 192.168.1.6 icmp_seq=4 ttl=64 time=5.140 ms
84 bytes from 192.168.1.6 icmp_seq=5 ttl=64 time=14.813 ms
```

PC6

```
PC6> ping 192.168.1.1
```

```
84 bytes from 192.168.1.1 icmp_seq=1 ttl=64 time=16.321 ms
84 bytes from 192.168.1.1 icmp_seq=2 ttl=64 time=24.786 ms
84 bytes from 192.168.1.1 icmp_seq=3 ttl=64 time=34.993 ms
84 bytes from 192.168.1.1 icmp_seq=4 ttl=64 time=19.314 ms
84 bytes from 192.168.1.1 icmp_seq=5 ttl=64 time=26.061 ms
```

```
PC6> ping 192.168.1.2
```

```
84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=14.662 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=21.941 ms
84 bytes from 192.168.1.2 icmp_seq=3 ttl=64 time=12.427 ms
```

```
84 bytes from 192.168.1.2 icmp_seq=4 ttl=64 time=31.497 ms
84 bytes from 192.168.1.2 icmp_seq=5 ttl=64 time=24.086 ms
```

```
PC6> ping 192.168.1.3
```

```
84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=25.078 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=15.737 ms
84 bytes from 192.168.1.3 icmp_seq=3 ttl=64 time=36.999 ms
84 bytes from 192.168.1.3 icmp_seq=4 ttl=64 time=44.908 ms
84 bytes from 192.168.1.3 icmp_seq=5 ttl=64 time=24.925 ms
```

```
PC6> ping 192.168.1.4
```

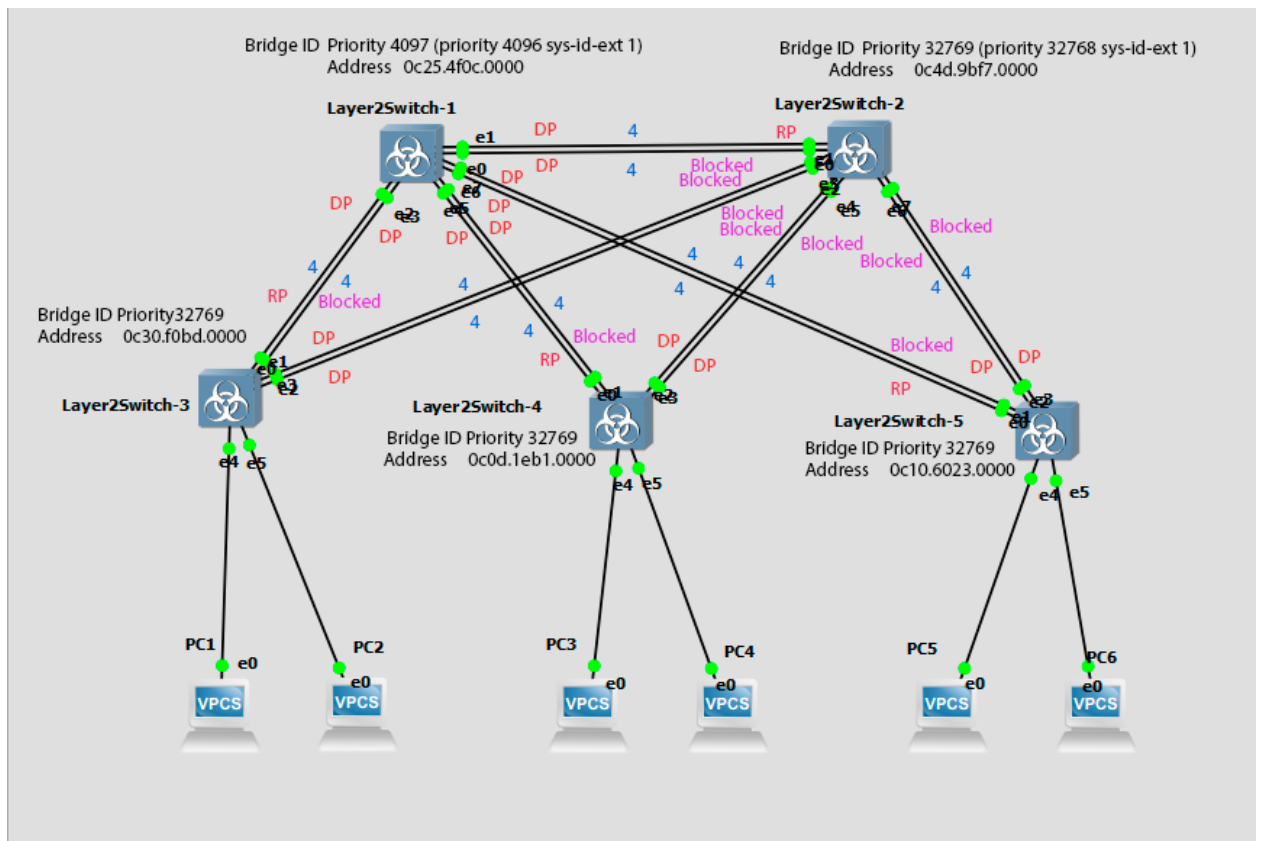
```
84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=30.930 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=17.348 ms
84 bytes from 192.168.1.4 icmp_seq=3 ttl=64 time=10.419 ms
84 bytes from 192.168.1.4 icmp_seq=4 ttl=64 time=14.205 ms
84 bytes from 192.168.1.4 icmp_seq=5 ttl=64 time=45.110 ms
```

```
PC6> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=7.215 ms
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=5.674 ms
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=14.364 ms
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=9.175 ms
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=15.180 ms
```

3) На изображении схемы отметить BID каждого коммутатора и режимы работы портов (RP/DP/blocked) и стоимости маршрутов, результат сохранить в файл.

Стоимость всех маршрутов: 4



4) При помощи Wireshark отследить передачу пакетов hello от корневого коммутатора на всех линках (nb!), результаты включить в отчет.

The screenshot shows a Wireshark capture of STP Hello packets. The packet list on the left shows several STP Configuration BPDUs (Type 0) from the root bridge (4096) to various destinations. The selected packet (No. 1361) is expanded in the packet details pane, showing the Spanning Tree Protocol fields: Protocol Identifier (0x0000), Protocol Version Identifier (0), BPDU Type (Configuration), BPDU flags (0x00), Root Identifier (4096), Root Path Cost (0), Bridge Identifier (4096), Port identifier (0x8002), Message Age (0), Max Age (20), Hello Time (2), and Forward Delay (15). The packet bytes pane shows the raw data of the BDU.

No.	Time	Source	Destination	Protocol	Length	Info
1349	486.328333	0c:4d:9b:f7:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8002
1350	486.431397	0c:25:4f:0c:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1351	487.661597	0c:4d:9b:f7:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8002
1352	487.839569	0c:25:4f:0c:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1353	487.848161	0c:25:4f:0c:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1354	488.762706	0c:4d:9b:f7:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8002
1355	489.084839	0c:25:4f:0c:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1356	489.426386	0c:25:4f:0c:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1357	489.699135	0c:25:4f:0c:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1358	489.824451	0c:4d:9b:f7:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8002
1359	490.011074	0c:4d:9b:f7:00:01	0c:4d:9b:f7:00:01	LOOP	60	Reply
1360	490.030903	0c:25:4f:0c:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1361	490.183909	0c:25:4f:0c:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1362	491.131170	0c:4d:9b:f7:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8002
1363	492.065034	0c:4d:9b:f7:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8002

The screenshot shows a Wireshark capture of STP Hello packets. The packet list on the left shows several STP Configuration BPDUs (Type 0) from the root bridge (4096) to various destinations. The selected packet (No. 5151) is expanded in the packet details pane, showing the Spanning Tree Protocol fields: Protocol Identifier (0x0000), Protocol Version Identifier (0), BPDU Type (Configuration), BPDU flags (0x00), Root Identifier (4096), Root Path Cost (0), Bridge Identifier (4096), Port identifier (0x8001), Message Age (0), Max Age (20), Hello Time (2), and Forward Delay (15). The packet bytes pane shows the raw data of the BDU.

No.	Time	Source	Destination	Protocol	Length	Info
5149	2071.035934	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001
5150	2072.309611	0c:4d:9b:f7:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8001
5151	2072.560260	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001
5152	2072.563555	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001
5153	2073.367845	0c:4d:9b:f7:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8001
5154	2073.482070	0c:25:4f:0c:00:00	0c:25:4f:0c:00:00	LOOP	60	Reply
5155	2073.955634	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001
5156	2074.297274	0c:4d:9b:f7:00:00	0c:4d:9b:f7:00:00	LOOP	60	Reply
5157	2074.298256	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001
5158	2074.393764	0c:4d:9b:f7:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8001
5159	2074.624977	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001
5160	2075.514229	0c:4d:9b:f7:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8001
5161	2075.814608	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001
5162	2075.819603	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001
5163	2076.561961	0c:4d:9b:f7:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8001
5164	2077.269522	0c:25:4f:0c:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8001

Standard input [Layer2Switch-1 Ethernet3 to Layer2Switch-3 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
1142	392.877253	0c:30:f0:bd:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8002
1143	392.896290	0c:25:4f:0c:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8004
1144	393.242942	0c:25:4f:0c:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8004
1145	393.910588	0c:30:f0:bd:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8002
1146	393.943628	0c:30:f0:bd:00:01	CDP/VTP/DTP/PagP/UD...	CDP	422	Device ID: vIOS-L2-01 Port ID: GigabitEthernet0/1
1147	394.476758	0c:25:4f:0c:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8004
1148	394.476795	0c:25:4f:0c:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8004
1149	394.855832	0c:30:f0:bd:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8002
1150	395.199543	0c:25:4f:0c:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8004
1151	395.567317	0c:30:f0:bd:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8002
1152	396.269861	0c:25:4f:0c:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8004
1153	396.525678	0c:30:f0:bd:00:01	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8002
1154	396.571094	0c:25:4f:0c:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8004

Frame 1148: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Port identifier: 0x8004

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

wireshark_Standard input874SU3.pcapng

Пакеты: 1154 - Потеряно: 0 (0.0%)

Профиль: Default

Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
5903	2235.045217	0c:30:f0:bd:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8001
5904	2235.923862	0c:25:4f:0c:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8003
5905	2236.050605	0c:25:4f:0c:00:02	0c:25:4f:0c:00:02	LOOP	60	Reply
5906	2236.069356	0c:30:f0:bd:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8001
5907	2236.215364	0c:25:4f:0c:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8003
5908	2237.233669	0c:30:f0:bd:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8001
5909	2237.283524	0c:25:4f:0c:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8003
5910	2237.871090	0c:25:4f:0c:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8003
5911	2237.873285	0c:25:4f:0c:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8003
5912	2238.108082	0c:30:f0:bd:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8001
5913	2238.828969	0c:30:f0:bd:00:00	0c:30:f0:bd:00:00	LOOP	60	Reply
5914	2239.120569	0c:30:f0:bd:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8001
5915	2239.232003	0c:25:4f:0c:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8003
5916	2239.634713	0c:25:4f:0c:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8003
5917	2240.206815	0c:30:f0:bd:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8001
5918	2240.246395	0c:25:4f:0c:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8003

Frame 5916: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Port identifier: 0x8003

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

wireshark_Standard input85UQU3.pcapng

Пакеты: 5921 - Потеряно: 0 (0.0%)

Профиль: Default

Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
6161	2321.560555	0c:25:4f:0c:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8005
6162	2322.466742	0c:0d:1e:b1:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8001
6163	2323.140578	0c:25:4f:0c:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8005
6164	2323.140611	0c:25:4f:0c:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8005
6165	2323.535814	0c:0d:1e:b1:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8001
6166	2323.739174	0c:25:4f:0c:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8005
6167	2324.234224	0c:0d:1e:b1:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8001
6168	2324.839878	0c:25:4f:0c:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8005
6169	2325.088979	0c:25:4f:0c:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8005
6170	2325.242653	0c:0d:1e:b1:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8001
6171	2325.306206	0c:25:4f:0c:00:04	COP/VTP/DTP/PAGP/UD...	DTP	62	Dynamic Trunk Protocol
6172	2326.477959	0c:0d:1e:b1:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8001
6173	2326.548219	0c:25:4f:0c:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8005
6174	2326.551010	0c:25:4f:0c:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8005

Frame 6174: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bit)

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Port Identifier: 0x8005

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

Spanning Tree Protocol: Protocol

Пакеты: 6174 · Потеряно: 0 (0.0%)

Профиль: Default

Standard input [Layer2Switch-1 Ethernet5 to Layer2Switch-4 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

Time	Source	Destination	Protocol	Length	Info
224	419.653029	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
225	419.785507	0c:0d:1e:b1:00:01	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/200/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8002
226	419.874927	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
227	420.701911	0c:0d:1e:b1:00:01	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/300/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8002
228	421.004824	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
229	421.009889	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
230	421.582953	0c:0d:1e:b1:00:01	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/100/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8002
231	422.158085	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
232	422.281751	0c:0d:1e:b1:00:01	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/200/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8002
233	422.516554	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
234	422.815229	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
235	423.235921	0c:0d:1e:b1:00:01	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/300/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8002
236	423.990026	0c:0d:1e:b1:00:01	0c:0d:1e:b1:00:01	LOOP	60 Reply
237	424.009529	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
238	424.102094	0c:25:4f:0c:00:05	Nearest-Customer-Br...	STP	60 Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8006
239	424.246099	0c:0d:1e:b1:00:01	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/100/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8002
240	424.907668	0c:0d:1e:b1:00:01	Nearest-Customer-Br...	STP	60 Conf. Root = 32768/200/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8002

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Port Identifier: 0x8006

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

wireshark_Standard inputKE0SU3.pcapng

Пакеты: 1240 · Потеряно: 0 (0.0%)

Профиль: Default

Standard input [Layer2Switch-1 Ethernet6 to Layer2Switch-5 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
6334	2383.888895	0c:25:4f:0c:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8007
6335	2384.872543	0c:25:4f:0c:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8007
6336	2385.004614	0c:10:60:23:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:10:60:23:00:00 Cost = 0 Port = 0x8001
6337	2385.085617	0c:25:4f:0c:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8007
6338	2385.271379	0c:25:4f:0c:00:06	0c:25:4f:0c:00:06	LOOP	60	Reply
6339	2385.626291	0c:10:60:23:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:10:60:23:00:00 Cost = 0 Port = 0x8001
6340	2386.562051	0c:25:4f:0c:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8007
6341	2386.563238	0c:25:4f:0c:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8007
6342	2386.726428	0c:10:60:23:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:10:60:23:00:00 Cost = 0 Port = 0x8001
6343	2387.120319	0c:25:4f:0c:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8007
6344	2387.444922	0c:10:60:23:00:00	0c:10:60:23:00:00	LOOP	60	Reply
6345	2387.934649	0c:10:60:23:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:10:60:23:00:00 Cost = 0 Port = 0x8001
6346	2388.269961	0c:25:4f:0c:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8007
6347	2388.490695	0c:25:4f:0c:00:06	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8007
6348	2388.614560	0c:10:60:23:00:00	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:10:60:23:00:00 Cost = 0 Port = 0x8001
6349	2389.573264	0c:25:4f:0c:00:06	0c:25:4f:0c:00:06	LOOP	60	Reply

Frame 6346: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Port identifier: 0x8007

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

wireshark_Standard input[HVI7U3.pcapng] Пакеты: 6352 · Потеряно: 0 (0.0%) Профиль: Default

Standard input [Layer2Switch-1 Ethernet7 to Layer2Switch-5 Ethernet1]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
1231	421.636336	0c:10:60:23:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:10:60:23:00:00 Cost = 0 Port = 0x8002
1232	421.971070	0c:25:4f:0c:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8008
1233	422.290304	0c:25:4f:0c:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8002
1234	422.580045	0c:10:60:23:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:10:60:23:00:00 Cost = 0 Port = 0x8002
1235	423.176104	0c:10:60:23:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:10:60:23:00:00 Cost = 0 Port = 0x8002
1236	423.603335	0c:25:4f:0c:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8008
1237	423.615839	0c:25:4f:0c:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8008
1238	424.334586	0c:25:4f:0c:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8008
1239	424.339814	0c:10:60:23:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/300/0c:10:60:23:00:00 Cost = 0 Port = 0x8002
1240	424.550650	0c:10:60:23:00:01	0c:10:60:23:00:01	LOOP	60	Reply
1241	425.327604	0c:25:4f:0c:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8008
1242	425.497580	0c:10:60:23:00:01	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/100/0c:10:60:23:00:00 Cost = 0 Port = 0x8002
1243	425.679082	0c:25:4f:0c:00:07	Nearest-Customer-Bridge	STP	60	Conf. Root = 32768/200/0c:25:4f:0c:00:00 Cost = 0 Port = 0x8008

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Bridge Priority: 4096

Root Bridge System ID Extension: 1

Root Bridge System ID: 0c:25:4f:0c:00:00 (0c:25:4f:0c:00:00)

Root Path Cost: 0

Bridge Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Port identifier: 0x8008

Message Age: 0

Max Age: 20

Hello Time: 2

Forward Delay: 15

Root Bridge System ID Extension (stp.root.ext), 2 byte(s) Пакеты: 1243 · Потеряно: 0 (0.0%) Профиль: Default

Standard input [Layer2Switch-3 Ethernet3 to Layer2Switch-2 Ethernet3]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
6633	2414.659860	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8004
6634	2415.186776	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8004
6635	2415.613156	0c:4d:9b:f7:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8004
6636	2416.208645	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8004
6637	2416.213117	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8004
6638	2416.620513	0c:4d:9b:f7:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8004
6639	2417.261029	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8004
6640	2417.646347	0c:4d:9b:f7:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8004
6641	2417.733137	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8004
6642	2418.257003	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8004
6643	2418.990524	0c:4d:9b:f7:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8004
6645	2419.348167	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8004
6646	2419.351482	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8004
6647	2420.091204	0c:4d:9b:f7:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8004
6648	2420.352580	0c:30:f0:bd:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8004

Frame 6646: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:30:f0:bd:00:00

Port identifier: 0x8004

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

Spanning Tree Protocol: Protocol

Пакеты: 6648 · Отображено: 6155 (92.6%) · Потеряно: 0 (0.0%) · Профиль: Default

Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
6625	2411.173072	0c:4d:9b:f7:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8003
6626	2411.624550	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8003
6627	2412.147902	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8003
6628	2412.574189	0c:4d:9b:f7:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8003
6629	2413.170223	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8003
6630	2413.178812	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8003
6631	2413.581378	0c:4d:9b:f7:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8003
6632	2414.222957	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8003
6633	2414.607610	0c:4d:9b:f7:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8003
6634	2414.699271	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8003
6635	2415.218197	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8003
6636	2415.951741	0c:4d:9b:f7:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8003
6637	2416.309408	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:30:f0:bd:00:00 Cost = 0 Port = 0x8003
6638	2416.317118	0c:30:f0:bd:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8003
6639	2417.051937	0c:4d:9b:f7:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8003

Frame 6638: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:30:f0:bd:00:00

Port identifier: 0x8003

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

Spanning Tree Protocol: Protocol

Пакеты: 6640 · Отображено: 6148 (92.6%) · Потеряно: 0 (0.0%) · Профиль: Default

Захват с Standard input [Layer2Switch-2 Ethernet4 to Layer2Switch-4 Ethernet2]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
8642	3129.034159	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8003
8643	3129.649430	0c:4d:9b:f7:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8005
8644	3129.769644	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8003
8646	3130.470709	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8003
8647	3130.740001	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8003
8648	3131.068206	0c:4d:9b:f7:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8005
8651	3131.958364	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8003
8652	3131.959593	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8003
8653	3132.280333	0c:4d:9b:f7:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8005
8654	3132.670677	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8003
8655	3133.349760	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8003
8656	3133.545201	0c:4d:9b:f7:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8005
8657	3133.600650	0c:0d:1e:b1:00:02	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8003

Frame 8655: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:0d:1e:b1:00:00

Port identifier: 0x8003

Message Age: 1

Max Age: 20

Hello Time: 2

Forward Delay: 15

Spanning Tree Protocol: Protocol

Пакеты: 8657 - Отображено: 8013 (92.6%)

Профиль: Default

Захват с Standard input [Layer2Switch-2 Ethernet5 to Layer2Switch-4 Ethernet3]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

stp

No.	Time	Source	Destination	Protocol	Length	Info
8633	3126.496320	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8004
8634	3127.112741	0c:4d:9b:f7:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8006
8635	3127.231954	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8004
8636	3127.232395	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8004
8637	3128.202280	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8004
8638	3128.530526	0c:4d:9b:f7:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8006
8639	3128.694797	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8004
8642	3129.421621	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/100/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8004
8643	3129.743044	0c:4d:9b:f7:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8006
8644	3130.133089	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/200/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8004
8645	3130.133217	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8004
8646	3131.009488	0c:4d:9b:f7:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:4d:9b:f7:00:00 Cost = 0 Port = 0x8006
8647	3131.062782	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/300/0c:0d:1e:b1:00:00 Cost = 0 Port = 0x8004
8648	3131.453725	0c:0d:1e:b1:00:03	Nearest-Customer-Br...	STP	60	Conf. Root = 4096/1/0c:25:4f:0c:00:00 Cost = 4 Port = 0x8004

Frame 8639: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on 0

IEEE 802.3 Ethernet

Logical-Link Control

Spanning Tree Protocol

Protocol Identifier: Spanning Tree Protocol (0x0000)

Protocol Version Identifier: Spanning Tree (0)

BPDU Type: Configuration (0x00)

BPDU flags: 0x00

Root Identifier: 4096 / 1 / 0c:25:4f:0c:00:00

Root Path Cost: 4

Bridge Identifier: 32768 / 1 / 0c:0d:1e:b1:00:00

Port identifier: 0x8004

Message Age: 1

Max Age: 20

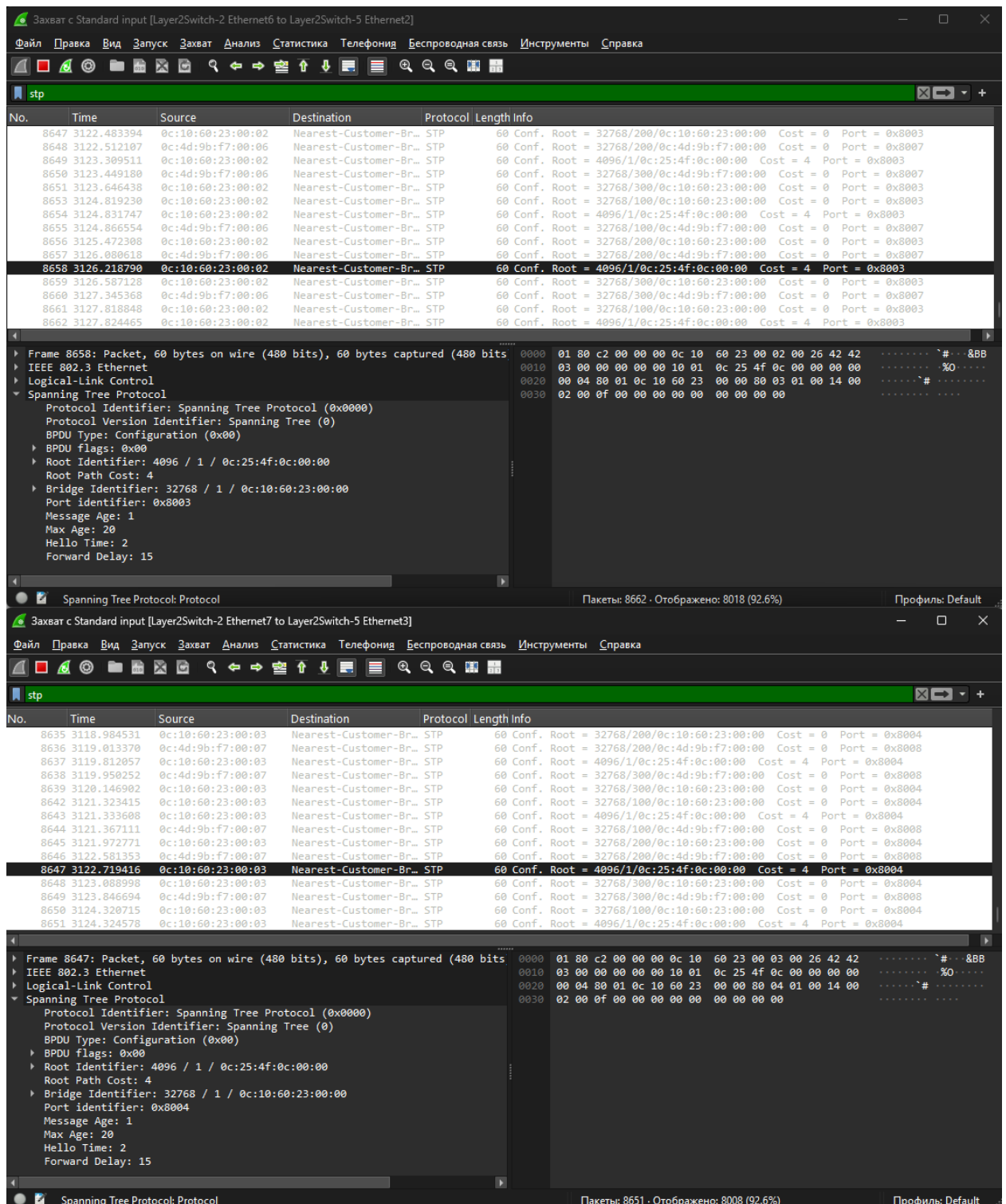
Hello Time: 2

Forward Delay: 15

Spanning Tree Protocol: Protocol

Пакеты: 8648 - Отображено: 8006 (92.6%)

Профиль: Default



Для фильтрации использовался фильтр stp. На всех линках были Configuration BPDUs, содержащие информацию о корневом коммутаторе. Для VLAN 1 в поле Root Identifier указан BID корневого коммутатора Layer2Switch-1 — 4096 / 1 / 0c:25:4f:c0:00:00. Значение Hello Time = 2 показывает, что служебные BPDUs формируются с периодом 2 секунды.

5) Изменить стоимость маршрута для порта RP произвольного назначенного (designated) коммутатора, повторить действия из п.3, результат сохранить в отдельный файл.

```

enable
configure terminal
interface GigabitEthernet0/0
spanning-tree vlan 1 cost 10
end

```

Layer2Switch-3 - PuTTY

```

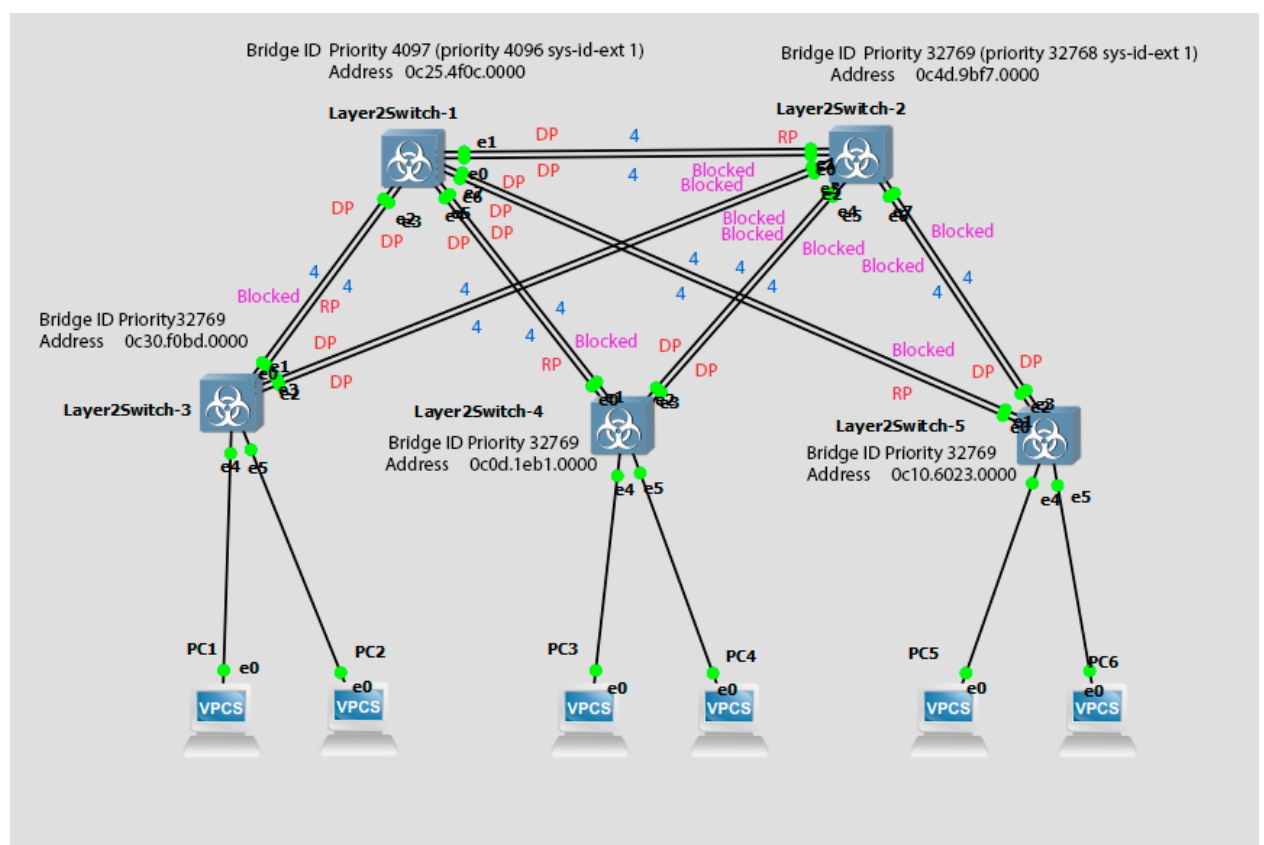
VLAN0001
Spanning tree enabled protocol ieee
Root ID    Priority    4097
           Address    0c25.4f0c.0000
           Cost       4
           Port       2 (GigabitEthernet0/1)
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
           Address    0c30.f0bd.0000
           Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time  15 sec

Interface   Role Sts Cost      Prio.Nbr Type
-----
Gi0/0       Altn BLK 10       128.1   Shr
Gi0/1       Root LRN 4        128.2   Shr
Gi0/2       Desg FWD 4        128.3   Shr
Gi0/3       Desg FWD 4        128.4   Shr
Gi1/0       Desg FWD 4        128.5   Shr
Gi1/1       Desg FWD 4        128.6   Shr

```

--More--



6) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств.